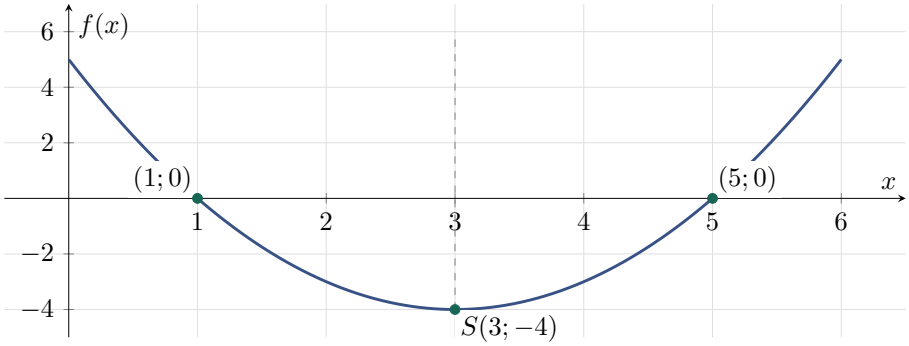
2.1 Three forms, three readings

Definition and results

A function of the second degree written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The developed form facilitates the calculation of $f(0)$; the canonical form $a(x-\alpha)^2 + \beta$ reveals summit and extremum; a factorized shape $a(x-r_1)(x-r_2)$ reveals roots and signs. Roots are the solutions of $f(x) = 0$.

Status and justification. The completion of the square is based on $x^2+2ux = (x+u)^2-u^2$. The existence of the canonical form is demonstrated by development; its general calculation is not an automatism payable.

Example solved. $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is -4 in 3 ; The roots are 1 and 5; the order originally is 5.



Curve of $f(x) = (x-3)^2 - 4$. The two roots and the top are indicated; the axis of symmetry is the right $x = 3$.

Point of vigilance

A factorized expression does not get by putting each of the terms separately between parentheses.

Immediate application. Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and sound minimum.

2.2 Discrimination and resolution

Definition and results

For $ax^2 + bx + c = 0$ with $a \neq 0$, we put it down $b^2 - 4ac$. If < 0, no real root; if 0, one root $-\frac{b}{2a}$; if > 0, two roots $\frac{-b - \sqrt{\Delta}}{2a}$ and $\frac{-b + \sqrt{\Delta}}{2a}$. To order roots, consider the sign of a.

Status and justification. By multiplying the equation by 4 $a \neq 0$, one obtains (2 $ax + b)^2$ One. square is positive or zero: the three cases follow. > 0, $2ax + b = \pm \sqrt{\Delta}$. Disjunction of cases avoids take the square root of a negative number.

Example solved. $2x^2 + x - 3 = 0 : \Delta = 1 + 24 = 25$; $x = \frac{-1 \pm 5}{4}$ i.e. 1 and $-\frac{3}{2}$. Verification: $(x - 1)(2x + 3) = 2x^2 + x - 3$.

Point of vigilance

The denominator is 2 a, not 2 ; the sign of b is taken with its parentheses in b^2.

Immediate application. Solve 3 $x^2 - 6x$ + 3 = 0 and $x^2 + 2x + 5 = 0$.

2.3 Factoring without automating the discriminant

Definition and results

If two real roots exist, their sum is $-\frac{b}{a}$ and their produce $\frac{c}{a}$. An apparent root or a remarkable identity often allows a shorter factorization. degree 2 cancelling in two distinct real r and s are exactly $a(x - r)(x - s)$ with $a \neq 0$.

Status and justification. Developing $a(x - r)(x - s) = ax^2 - a(r + s)x + ars$, identification the coefficients give the relationships. Conversely, if r and s have this sum and this product, the development regains the polynomial.

Example solved. $x^2 - 7x$ + 12 : 3 and 4 have sum 7 and produce 12. $x - 3)(x -$ 4. For to impose more $f(0)$ = 24, it takes 12 a = 24, so $a = 2$.

Point of vigilance

The sum of the roots is $-\frac{b}{a}$; don't forget the sign less. Only one root is not enough to determine a.

Immediate application. Determine the level 2 f polynomial such that $f(1) = f(4) = 0$ and $f(0) = -8$.

2.4 Sign, inequities and optimization

Definition and results

For two distinct roots $r < s$, the trinomial has the sign of a outside [$r; s$] and the sign between the roots. It is zero to the roots. Without real root, it keeps the sign of a. With a double root, it keeps the sign of a out of it. The optimization is done on the domain the problem, which may limit access to the summit.

Status and justification. The sign table of factors $a(x - r)(x - s)$ gives the result. canonical form compares $f(x)$ à β with the sign of $a(x - \alpha)^2$.

Example solved. $-2(x - 1)(x - 5) \geq 0$ equals $x \in$ [1; 5]. $A(x) = x(12 - x)$ on [0; 12], $A(x) = 36 - (x - 6)^2 \leq 36$, with equality in 6. A rectangle of perimeter 24 m has therefore an area maximum of 36 m ^{2} when it is square.

Point of vigilance

In a strict inequation, the roots are excluded. Check the geometric bounds before conclude.

Immediate application. Solve $x^2 - 4x - 5 < 0$ then maximize $x(8 - x)$ on [0; 3].